Standby generator

AN-002
HOW IT WORKS
2,500 KVA DIESEL STANDBY SET

What happens when a data centre loses its mains power?

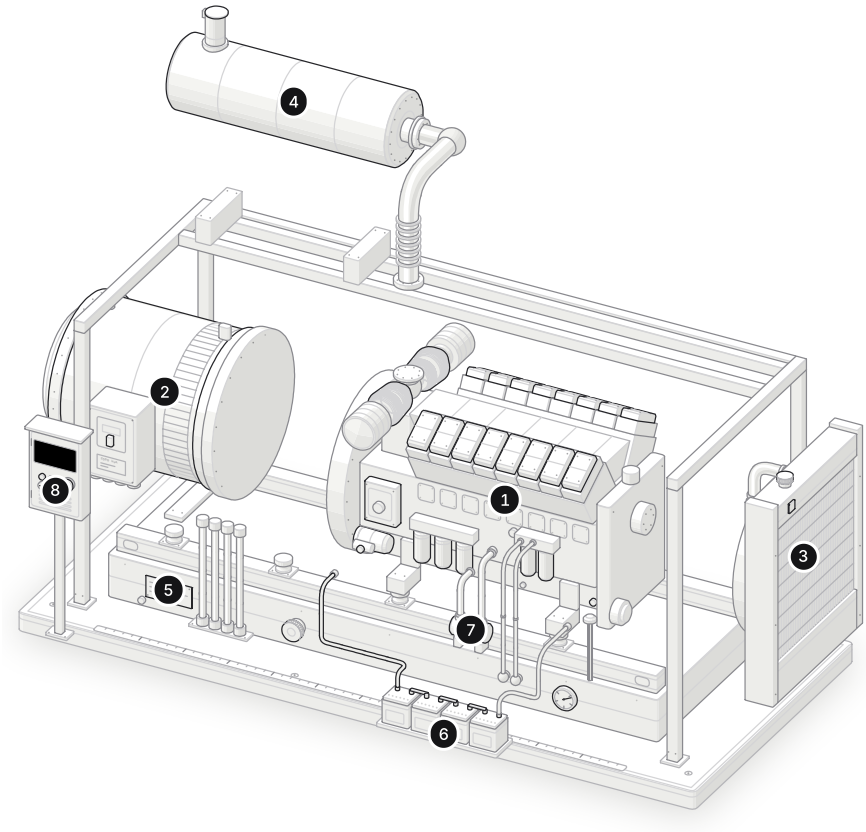
The UPS batteries carry the servers for the first seconds. Within 8.8 s the diesel generator has cranked, fired, reached 1,500 rpm and 50 Hz, and the transfer switch has moved the load across to it.

2,500 kVA
2,000 KW AT 0.8 POWER FACTOR

1,500 rpm
16-CYLINDER V DIESEL, 50 HZ

8.8 s
MAINS FAILURE TO ON LOAD

5,500 L
BASE TANK, 13 H AT 1,600 KW



01 **Diesel engine**
V16, twin turbos; turns the alternator at 1,500 rpm.

02 **Alternator**
Makes 400 V, 50 Hz power; breaker and cables in its terminal box.

03 **Radiator and fan**
Sheds the engine's heat to the air of the generator hall.

04 **Exhaust silencer**
Quietens the exhaust; the rain cap lifts when gas flows.

05 **Base fuel tank and mounts**
5,500 L of diesel under the skid; rubber mounts stop vibration.

06 **Starter batteries**
Four 12 V blocks give 24 V to crank the engine.

07 **Jacket-water heater**
Keeps the coolant at 40 °C so the set starts fast.

08 **Control panel**
Senses the mains failure, starts, watches and protects the set.